Exchange rate and Production network

Macro Internal Seminar

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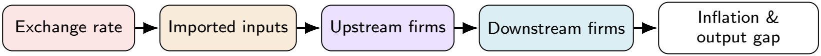
Motivation



Firms buy inputs mostly from *other domestic firms*, not just labor — a cost shock to one sector mechanically hits everyone downstream

Imported inputs are the norm (Chile: up to 19.5% in Manufacturing) — so an FX move is, for the firm, a domestic cost shock

Network models get this propagation right but ignore the exchange rate; open-economy models get the exchange rate right but ignore the network

This paper: both together

Does accounting for production networks change the optimal exchange-rate regime?

Literature

Network side (closed economy): Rubbo (2024); La'O & Tahbaz-Salehi (2022); Pasten, Schoenle & Weber (2020); Ozdagli & Weber (2026); Ghassibe (2021)

Open-economy side (no network): Galí & Monacelli (2005); Fanelli & Straub (2021); Schmitt-Grohé & Uribe (2003); Gopinath & Itskhoki (2022); Amiti et al. (2019); Auer, Levchenko & Sauré (2019); Itskhoki & Mukhin (2023); Bianchi & Coulibaly (2025); Coulibaly (2023)

Open economy with network (closest to this paper):

- **Gnocato, Montes-Galdón & Stamato (2025, ECB); Kalemli-Özcan et al. (2025)** — open economy *with* network (tariffs), fixed Taylor rule, no FX-regime choice[1]
- **Qiu, Wang, Xu & Zanetti (2026, JME)** — closest paper: derives the open-economy DC index directly extending Rubbo, WIOD 43-country calibration; but *static* (no NFA/persistence), no UIP, and no FX-regime choice

Gap: network + open economy + FX regime choice, together.

[1] Details & what these papers cite for open-economy production networks: Appendix.

This Paper

Extend Rubbo $\rightarrow$ SOE: imported inputs Ω^F, UIP, debt-elastic premium

Ask: optimal FX regime? DC index survive with 2 cost-push channels?

Three literatures, none overlapping fully (network only; open economy only; network + open economy but no regime choice) — **first to put all three together**

Not just Galí-Monacelli with sectors: ranking survives, but the mechanism (risk-premium) and sector heterogeneity are new

Calibrate: Chile's real I-O table (Banco Central), also Korea, Czechia & a stylized baseline; solve **nonlinear** model directly

Households

Small open economy: home takes foreign price P_t^{F*}, rate i^*, and demand D_t^* as given.

Representative household chooses $\{C_t, L_t, B_t, B_t^*\}$ to maximize lifetime utility, subject to a budget constraint with a domestic bond B_t and a foreign bond B_t^*:

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\varphi}}{1+\varphi} \right]$$

$$P_t^C C_t + B_t + S_t B_t^* = W_t L_t + \Pi_t + (1+i_{t-1}) B_{t-1} + (1+i^*) \left[1 - \psi(\hat{b}_{t-1}^* - \bar{b}^*)\right] S_t B_{t-1}^*$$

Consumption is CES across home/foreign goods, Cobb-Douglas across domestic sectors:

$$C_t = \left[\omega^{1/\eta} (C_t^H)^{\frac{\eta-1}{\eta}} + (1-\omega)^{1/\eta} (C_t^F)^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}, \qquad C_t^H = \prod_i (C_{it}^H)^{\beta_i^H}$$

Firms: Technology

Firms in sector i share Cobb-Douglas technology (labor, domestic inputs, imports) and price under monopolistic competition; CRS makes marginal cost common across firms:

$$Y_{ift} = A_{it}\, L_{ift}^{\alpha_i} \prod_j X_{ijft}^{\Omega_{ij}^H}\, M_{ift}^{\Omega_i^F}$$

$$MC_{it} = \frac{W_t^{\alpha_i} \prod_j P_{jt}^{\Omega_{ij}^H} \, (S_t P_t^{F*})^{\Omega_i^F}}{A_{it}}$$

Two cost-push channels: MC_{it} moves with A_{it} (TFP) *and* S_t, the exchange rate (via the import cost $S_t P_t^{F*}$); with **Calvo pricing**, only a fraction δ_i resets each period.

Firms: Production Network

Stacking each firm's input shares Ω^H_{ij}, Ω^F_i across sectors gives the **input-output network** $\Omega^H \in \mathbb{R}^{N\times N}$, import-exposure vector $\boldsymbol{\omega}_F \in \mathbb{R}^N$ (import cost share of each sector's output), and export-exposure vector $\boldsymbol{\omega}_X \in \mathbb{R}^N$ (export share of each sector's own output).

The **Leontief inverse** sums direct and indirect linkages:

$$(I - \Omega^H)^{-1} = I + \Omega^H + (\Omega^H)^2 + \cdots$$

applied to consumption shares $\boldsymbol{\beta}^H$ (Households) it gives each sector's **Domar weight**; applied to $\boldsymbol{\omega}_F$ or $\boldsymbol{\omega}_X$, its import or export centrality:

$$\boldsymbol{\lambda}_D^\top = (\boldsymbol{\beta}^H)^\top (I - \Omega^H)^{-1}, \qquad \mathcal{M} = (I - \Omega^H)^{-1}\boldsymbol{\omega}_F, \qquad \mathcal{M}^X = (I - \Omega^H)^{-1}\boldsymbol{\omega}_X$$

Firms: Price Setting

Calvo pricing through the network delivers the (vector) Phillips curve:

$$\boldsymbol{\pi}_t = \rho(I - \mathcal{V})\mathbb{E}_t \boldsymbol{\pi}_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\boldsymbol{\chi}_t + \Gamma \Delta e_t$$

Same amplifier, two shocks: $\mathcal{B}, \Gamma$ are the *same* rigidity-adjusted operator $\tilde{A} \equiv \Delta(I - \Omega^H \Delta)^{-1}$, applied to labor share $\boldsymbol{\alpha}$ vs. import share $\boldsymbol{\omega}_F$:

$$\mathcal{B} = \frac{\tilde{A}\boldsymbol{\alpha}}{\kappa}, \qquad \Gamma = \frac{\tilde{A}\boldsymbol{\omega}_F}{\kappa}$$

Only relative shocks matter: $\mathcal{V}\mathbf{1} = \mathbf{0}$ — uniform TFP shocks create no cost-push; only sector-specific gaps $\boldsymbol{\chi}_t = (I - \Omega^H)^{-1} \log \mathbf{A}_t - \boldsymbol{p}_{t-1}$ do.

where:

$\boldsymbol{\pi}_t, \tilde{y}_t$: infl., gap	$\boldsymbol{\chi}_t$: TFP cost-push	Δe_t: exch. rate chg.	κ: GE scalar
$\mathcal{B}, \Gamma$: PC slope, FX pass-thru	$\mathcal{V}$: TFP cost-push matrix	$\tilde{A}$: rigidity-adj. Leontief	

Foreign Sector

Law of one price and UIP with a debt-elastic risk premium RP_t:

$$P_t^{FH} = S_t P_t^{F*}, \qquad i_t = i^* + \mathbb{E}_t \Delta e_{t+1} + \psi(\hat{b}_t^*) + RP_t$$

NFA accumulates the trade balance; exports are **sector-specific** (Resource-heavy, matching Chile's data):

$$\hat{b}_t^* = \frac{1 + i_{t-1}}{\Pi_t^C} \hat{b}_{t-1}^* + \frac{P_t^H \sum_i EX_{it} - P_t^{FH} IM_t}{P_t^C}, \qquad EX_{it} = \kappa_{EX,i} D_t^* \, PX_t \left(\frac{P_{it}}{P_t^{FH}} \right)^{-\theta^*}$$

Two channels to the domestic economy:

$$\hat{s}_t + \hat{p}_t^{F*} \ \rightarrow \ MC_{it} \ \rightarrow \ \pi_{it} \quad \text{(cost-push)} \qquad \hat{s}_t \ \rightarrow \ EX_{it} \ \rightarrow \ \tilde{y}_t \ \rightarrow \ \pi_{it} \quad \text{(demand)}$$

where:

P_t^{F*}: foreign price of imports RP_t: risk-premium shock D_t^*: foreign demand shock PX_t: export price/ToT shock

$\psi(\cdot)$: debt-elastic NFA stabilizer $\hat{b}_t^*$: NFA (% GDP) $\kappa_{EX,i}$: export scale, sector i θ^*: export demand elasticity

$\hat{s}_t \equiv \log(S_t/\bar{S})$: log exch. rate $\Delta e_{t+1} \equiv \mathbb{E}_t \hat{s}_{t+1} - \hat{s}_t$: exp. depreciation   (hats: log-deviation from steady state, throughout)

Monetary Policy Regimes

Three exchange-rate regimes are compared.

Free float

$$\hat{i}_t = \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t$$

The exchange rate adjusts endogenously.

Hard peg

$$\hat{s}_t = 0$$

Adjustment occurs through domestic inflation and output.

Managed float

$$\hat{i}_t = \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \hat{s}_t$$

The central bank partially stabilizes the exchange rate.

Equilibrium System

Phillips curve:

$$\boldsymbol{\pi}_t = \rho(I - \mathcal{V})\mathbb{E}_t\boldsymbol{\pi}_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\boldsymbol{\chi}_t + \Gamma\Delta e_t$$

IS curve:

$$\tilde{y}_t = \mathbb{E}_t\tilde{y}_{t+1} - \frac{1}{\gamma}\left(i_t - \mathbb{E}_t\pi_t^C - r_t^{\mathrm{nat}}\right) - \frac{\eta(1-\omega)}{\gamma}\mathbb{E}_t\Delta\left(\hat{s}_t + \hat{p}_t^{F*} - \hat{p}_t^H\right)$$

Foreign sector:

$$i_t = i^* + \mathbb{E}_t\Delta e_{t+1} + \psi(\hat{b}_t^*) + RP_t$$

Policy rule:

$$\hat{i}_t = \begin{cases} \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t & \text{Float} \\ \text{residual } (\hat{s}_t = 0) & \text{Peg} \\ \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \hat{s}_t & \text{Managed} \end{cases}$$

where: r_t^{nat} natural real rate, $\hat{p}_t^H$ log home price idx., π_t^{DC} DC index (target); $\mathcal{B}, \Gamma, \mathcal{V}, \psi(\cdot), RP_t, \hat{p}_t^{F*}(PF_t)$ as before.

Welfare loss: $\mathcal{W} = \frac{1}{2}\left[(\gamma+\varphi)\mathrm{Var}(\tilde{y}_t) + \sum_i \kappa_i \mathrm{Var}(\pi_{it})\right]$, $\kappa_i = \lambda_{D,i}\varepsilon\frac{1-\hat{\delta}_i}{\hat{\delta}_i}$

Seven AR(1) shocks (1% s.d. each): sector TFP A_1, A_2, A_3 (ρ=0.90), import price PF (ρ=0.85), foreign demand D^* (ρ=0.80), export price/ToT PX (ρ=0.85), risk premium RP (ρ=0.80).

Calibration: Chile Data

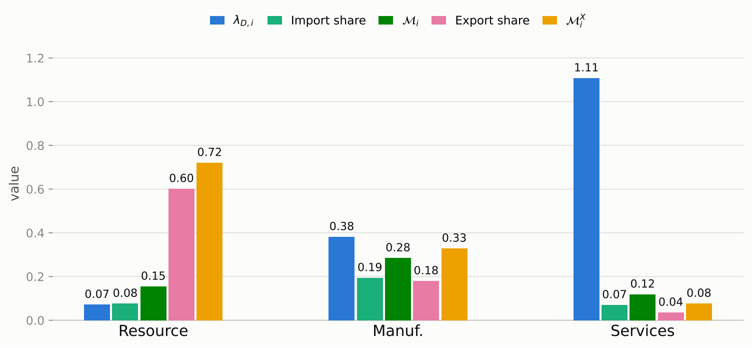
Parameter	Value	Source
$\alpha_1, \alpha_2, \alpha_3$	0.503, 0.359, 0.604	Data: BCCh IO table
$\Omega_1^F, \Omega_2^F, \Omega_3^F$	0.077, 0.195, 0.070	
$\beta_1^H, \beta_2^H, \beta_3^H$	0.027, 0.229, 0.744	
Export share$_{1,2,3}$	0.602, 0.180, 0.036	
$\delta_1, \delta_2, \delta_3$	0.90, 0.31, 0.16	Lit.: Dhyne et al. (2005); Nakamura–Steinsson (2008)
ψ	0.020	Key mechanism: risk-premium/UIP elasticity & FX-regime rules (see ψ-sensitivity, appendix)
ϕ_π	1.50	
ϕ_y	0.50	
ϕ_s	0.30	

Ω^H (order: Res., Man., Serv.; source: BCCh IO table):

$$\Omega^H = \begin{pmatrix} 0.075 & 0.153 & 0.193 \\ 0.099 & 0.202 & 0.145 \\ 0.002 & 0.058 & 0.266 \end{pmatrix}$$

Standard literature values (not paper-specific; sources in Appendix): β=0.99, γ=1.00, φ=2.00, η=1.50, ε=8.00, θ^*=2.00, $\beta^{F,tot}$=0.10.

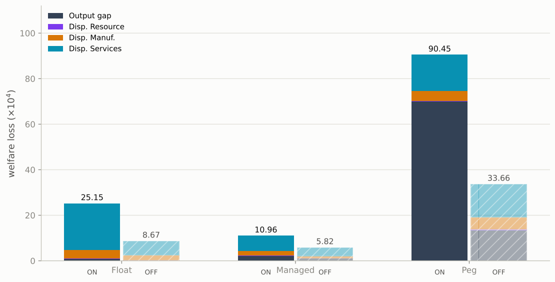
Network Properties: Import & Export Exposure



The production network amplifies both import and export exposure well beyond each sector's direct (raw) share — e.g. import exposure roughly doubles for every sector once indirect, network-propagated exposure is counted, not just each sector's own imports.

$\lambda_{D,i} = [(\beta^H)^\top (I - \Omega^H)^{-1}]_i$ (Domar weight), $\mathcal{M}_i = [(I - \Omega^H)^{-1} \boldsymbol{\omega}_F]_i$ and $\mathcal{M}_i^X = [(I - \Omega^H)^{-1} \boldsymbol{\omega}_X]_i$ (import/export centrality) — full notation: Appendix.

Welfare Ranking: Managed Float Dominates

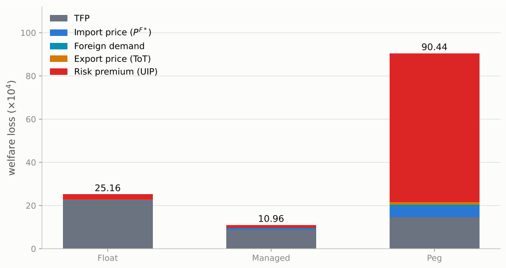


Managed $<$ Float $\ll$ Peg: 10.96/25.15/90.45 ($\times 10^{-4}$)

Float & Managed: almost entirely price dispersion, concentrated in Services

Without the network (hatched bars, $\Omega^H \equiv 0$): losses fall to 5.82/8.67/33.66 — **same ranking**, but the network roughly **doubles-to-triples** every regime's loss (already visible here; full ON/OFF comparison next slide)

What Drives Each Regime's Loss? Shock Decomposition



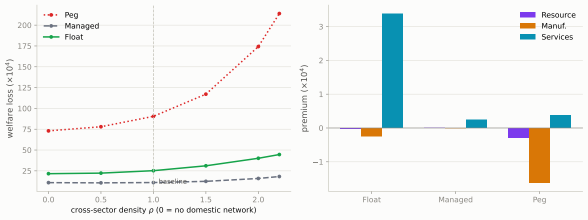
Peg: risk-premium/UIP $\rightarrow$ 76% of loss, mostly output gap — the shock that decides everything for a hard peg

Float & Managed: TFP shocks dominate instead (22.44 and 8.68 of their totals) — risk-premium barely registers (2.38/1.25) since the exchange rate (Float) or partial smoothing (Managed) absorbs it before it reaches output

Import price, foreign demand, and export price/ToT are second-order for every regime — the ranking is not a terms-of-trade story

Isolating the Network Channel

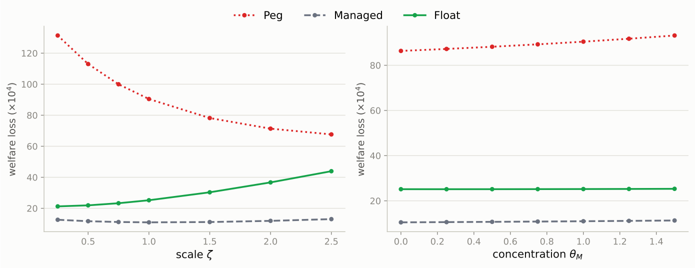
Beyond ON/OFF (Welfare Ranking, previous slides): does the network's effect scale *continuously* with how interconnected the economy is?



Density sweep on the **real Chile network** (scales all six cross-sector Ω^H entries by ρ, own-input fixed; ρ=1 baseline, max feasible $\rho \approx 2.2$): losses rise monotonically with density in every regime, and the Peg/Float gap **widens — Managed stays best throughout**. Right: price-dispersion premium (ρ=1 vs. ρ=0) is clearly positive and concentrated in Services for Float/Managed, but small and *mixed* across sectors for Peg (Resource/Manuf. slightly negative, Services slightly positive, netting close to zero) — Peg's density sensitivity runs mainly through the output gap, not price dispersion.

Import Openness & Import Concentration

Sweep import openness ζ and cross-sector concentration θ_M of $\boldsymbol{\omega}_F$, real Chile calibration.



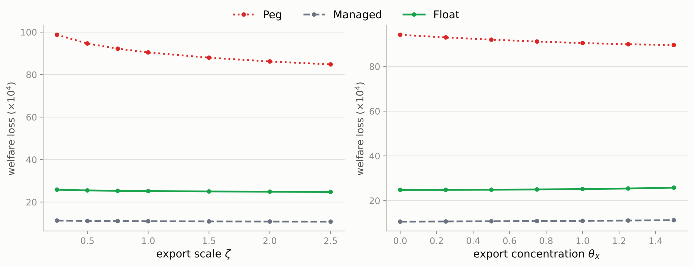
Managed $<$ Float $<$ Peg robust throughout

Openness ↑: Peg ↓, Float ↑ — gap **converges**

Concentration ↑: Peg ↑ faster — gap **diverges**

Export Openness & Export Concentration

Same construction, applied to export openness ζ and concentration θ_X of $\kappa_{EX,i}$, real Chile calibration.



Managed < Float < Peg robust throughout

Openness ↑: **all regimes improve**, Peg most

Concentration toward Resource (flexible sector): helps Peg, slightly hurts Float/Managed

Mechanism: How the Network Amplifies Shocks

One-line takeaway: the network **dampens** the inflation peak but **amplifies** the output-gap peak — in *every* regime, for *both* shocks. Cost-push isn't eliminated, it's **redirected**: spread into a persistent quantity distortion instead of a sharp price spike.

Shock	**Regime**	**Inflation peak**	**Output-gap peak**
		(Network ON ÷ Network OFF)	
Import-price	Float	0.40×	2.77×
	Managed	0.61×	2.28×
	Peg	0.86×	**3.15**× (worst)
Risk-premium	Float	0.41×	**2.57**× (worst)
	Managed	0.57×	0.98× (*no amplification*)
	Peg	0.67×	2.16×

Reading it: $< 1\times$ = network shrinks the peak; $> 1\times$ = network amplifies it. (Peak IRF response over 20 quarters, real Chile calibration; exchange-rate peak itself barely moves with the network, 0.8–1.5× — omitted above; full IRFs: Appendix.)

Import-price shock hits **Peg** hardest (no FX buffer); risk-premium hits **Float** hardest — **Managed** blocks the risk-premium shock's amplification almost entirely

Network Exposure & Sector Welfare Preferences

Different sectors prefer different regimes — sectors differ in stickiness $\hat{\delta}_i$ & network centrality $\lambda_{D,i}$, so κ_i and output-gap exposure hit them differently. Each cell: **Network ON / Network OFF**.

Price-dispersion cost ($\times 10^4$)

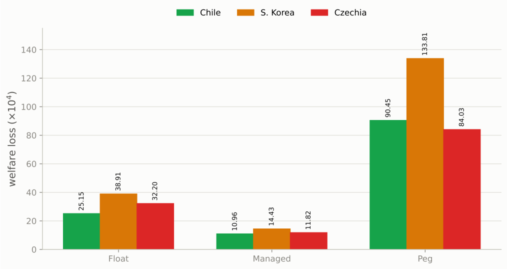
Sector	Float	Managed	Peg
Resource	0.05 / 0.03	0.04 / 0.02	0.27 / 0.27
Manuf.	3.87 / 2.23	1.92 / 0.94	4.28 / 5.32
Services	**20.25** / 6.22	6.71 / 3.74	**15.88** / 14.59

Var(own output gap) ($\times 10^4$)

Sector	Float	Managed	Peg
Resource	0.13 / 0.02	0.14 / 0.04	0.20 / 0.12
Manuf.	0.58 / 0.16	0.38 / 0.08	1.98 / 0.23
Services	0.38 / 0.15	2.31 / 1.33	**35.16** / 8.97

Price dispersion: Managed best everywhere, ON or OFF; Float's Services cost is mostly network-driven (20.25 ON vs. 6.22 OFF) — but **Peg's is not** (15.88 vs. 14.59, barely moves)
Output gap: Peg's Services outlier is overwhelmingly **network-driven** (35.16 ON vs. 8.97 OFF, +292%) — without the network Peg would still be worst, but nowhere near this severe

Robustness: South Korea & Czechia



Managed $<$ Float $\ll$ Peg in every calibration

UIP dominates Peg's loss throughout (65–76%)

Korea higher throughout: denser network & higher import exposure

Robustness Summary: What Drives the Ranking?

Managed < Float < Peg survives every sweep. What changes is the *size of the stakes* and *who bears them*:

Dimension	Effect on the regimes	Ranking?
Import openness ↑	Peg ↓↓ (131→68), Float ↑ (21→44), Managed flat (11–13)	gap **converges**
Import concentration ↑	Peg ↑↑ (86→93), Float/Managed roughly flat	gap **diverges**
Export openness ↑	**all regimes improve**, Peg most (99→85)	unchanged
Export concentration (Resource)	helps Peg, mildly hurts Float/Managed	unchanged
Network density ρ ↑	**all** worsen, Peg fastest (73→214 at ρ=2.2)	gap **diverges**

Import side drives the ranking; export side barely does — consistent with the mechanism running through cost-push ($\Gamma\Delta e_t$), not terms-of-trade/expenditure-switching

Volume vs. concentration point in opposite directions for Peg — more imports overall narrows the gap, but concentrating them in one sector widens it: it's exposure *structure*, not import *intensity*, that's dangerous

Density is the one dimension where every regime worsens, but unevenly — Peg's loss grows fastest, so a more interconnected economy specifically strengthens the case against pegging

A sector's own trade share is not what predicts its pain: Services has the *lowest* direct import share (7.0%) but the *largest* welfare cost — it sits atop the Domar-weight chain ($\lambda_{D,\text{Services}}$=1.11) and inherits shocks from every sector upstream; network-off cuts its Peg output-gap cost by 75% (35.16→8.97)

Conclusion

Networks change how the exchange rate operates: FX pass-through and TFP cost-push travel through the *same* rigidity-adjusted network operator — once S_t is a second cost-push channel, the closed-economy divine-coincidence index generically fails

Networks matter for severity and incidence, not the ranking: switching the domestic network on substantially raises welfare losses in every regime and reallocates which sector bears them — but **Managed $<$ Float $<$ Peg either way**

Peg is dominated by a different mechanism than the textbook story: not terms-of-trade, but risk-premium/UIP — zero monetary autonomy forces the shock straight into the output gap

Managed float wins because it is the only regime that dampens the FX pass-through channel without fully reimposing peg's loss of autonomy

Robust across three real-data calibrations (Chile, Korea, Czechia) and a genuine second-order solution